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N+ NETWORK FUNDAMENTALS BUILDING NETWORK

UNDERSTANDING AND MAINTANCE OF NETWORK

DAY5

TASKS:1

Connect your laptop and a friend's phone to the same Wi-Fi network, then use the ping command to check if both devices can communicate with each other. On Windows, use 'ping [IP address]' in Command Prompt; on Android, try a ping app or use Termux.

RESULT

1. Turn on the Wi-Fi on both the laptop and the friend's mobile phone.
2. Connect both devices to the **same Wi-Fi network**.
3. On the mobile phone, open **Settings → Wi-Fi → Connected Network** and note down the **IP Address**.
4. On the laptop, press **Windows + R**.
5. Type **cmd** and press **Enter** to open **Command Prompt**.
6. In the Command Prompt, type the following command: ping [phone’s ip adress]
7. Press **Enter**.
8. Wait for the ping process to complete and observe the results.

TASKS:2

Run the traceroute (tracert on Windows or traceroute on Mac/Linux) command to www.youtube.com from your device and note down the number of hops it takes to reach the destination.

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RESULT

1. 1.Open command prompt
2. 2.TYPE: tracert www.youtube.com
3. 3.Press entre
4. 4.Wait until command is complete
5. 5.Count the number of hops is displays

TASKS:3

Disconnect one device from the Wi-Fi, then try pinging it again from your laptop and describe the error message you receive.

RESULT

1. Connected the laptop and mobile phone to the same Wi-Fi network.
2. Verified that both devices could communicate using the **ping** command.
3. Turned **OFF the Wi-Fi** on the mobile phone.
4. Opened **Command Prompt** on the laptop.
5. Executed the following command: ping 192.168.29.122(EXAMPLE)
6. Observe the error occurs on the screen

TASKS:4

Create a simple table listing the IP addresses of all devices connected to your home Wi-Fi (use 'arp -a' on Windows), and identify which device belongs to which person or gadget.

RESULT

1. Connect the laptop to the home Wi-Fi network.
2. Press **Windows + R**.
3. Type **cmd** and press **Enter** to open Command Prompt.
4. Type the following command: arp -a
5. Press **Enter**.
6. Observe the list of IP addresses and Physical (MAC) addresses displayed.
7. Identify the devices connected to the Wi-Fi network, such as the router, laptop, mobile phone, smart TV, etc.
8. Create a table showing the IP address of each device and its corresponding device name.

TASKS:5

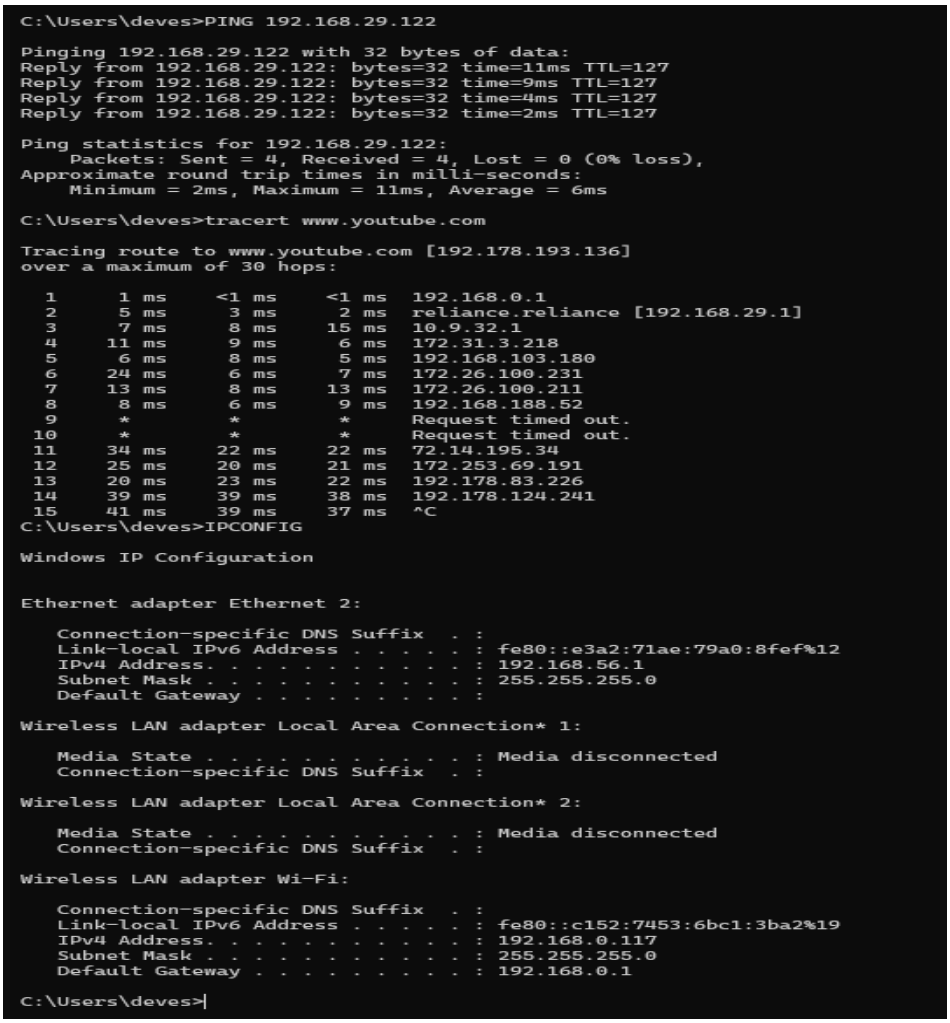
Write a one-page report explaining step-by-step how you set up the network, tested connectivity using ping and traceroute, and what results or issues you observed.Include screenshots or command outputs if possible.

RESULT

First, the laptop and a mobile phone were connected to the same Wi-Fi network. The IP address of the mobile phone was identified, and the **ping** command was executed from the laptop to test communication between the two devices. The ping test returned successful replies with **0% packet loss**, confirming that both devices were connected and could communicate properly.

Next, the **tracert** www.youtube.com command was executed to trace the route from the laptop to YouTube. The command displayed multiple network hops between the local router and the destination server. Although a few intermediate hops showed **"Request timed out"**, the traceroute reached the destination successfully, which is normal because some routers do not respond to traceroute requests.

Finally, the **ipconfig** command was used to verify the laptop's network configuration. The output showed the laptop's **IPv4 Address (192.168.0.117)**, **Subnet Mask (255.255.255.0)**, and **Default Gateway (192.168.0.1)**, confirming that the laptop was properly connected to the local Wi-Fi network.



```
C:\Users\deves>PING 192.168.29.122

Pinging 192.168.29.122 with 32 bytes of data:
Reply from 192.168.29.122: bytes=32 time=11ms TTL=127
Reply from 192.168.29.122: bytes=32 time=9ms TTL=127
Reply from 192.168.29.122: bytes=32 time=4ms TTL=127
Reply from 192.168.29.122: bytes=32 time=2ms TTL=127

Ping statistics for 192.168.29.122:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 2ms, Maximum = 11ms, Average = 6ms

C:\Users\deves>tracert www.youtube.com

Tracing route to www.youtube.com [192.178.193.136]
over a maximum of 30 hops:

  1     1 ms    <1 ms    <1 ms  192.168.0.1
  2     5 ms     3 ms     2 ms  reliance.reliance [192.168.29.1]
  3     7 ms     8 ms    15 ms  10.9.32.1
  4    11 ms     9 ms     6 ms  172.31.3.218
  5     6 ms     8 ms     5 ms  192.168.103.180
  6    24 ms     6 ms     7 ms  172.26.100.231
  7    13 ms     8 ms    13 ms  172.26.100.211
  8     8 ms     6 ms     9 ms  192.168.188.52
  9     *        *        *     Request timed out.
 10     *        *        *     Request timed out.
 11    34 ms    22 ms    22 ms  72.14.195.34
 12    25 ms    20 ms    21 ms  172.253.69.191
 13    20 ms    23 ms    22 ms  192.178.83.226
 14    39 ms    39 ms    38 ms  192.178.124.241
 15    41 ms    39 ms    37 ms  ^C
C:\Users\deves>IPCONFIG

Windows IP Configuration


Ethernet adapter Ethernet 2:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::e3a2:71ae:79a0:8fef%12
   IPv4 Address. . . . . . . . . . . : 192.168.56.1
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :

Wireless LAN adapter Local Area Connection* 1:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Local Area Connection* 2:

   Media State . . . . . . . . . . . : Media disconnected
   Connection-specific DNS Suffix  . :

Wireless LAN adapter Wi-Fi:

   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::c152:7453:6bc1:3ba2%19
   IPv4 Address. . . . . . . . . . . : 192.168.0.117
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : 192.168.0.1

C:\Users\deves>
```

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